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3 Participation, personal development, and performance through youth sport

Jean Côté, Leisha Strachan, and Jessica Fraser-Thomas

Introduction

Youth sport has the potential to accomplish three important objectives in children's development (Côté and Fraser-Thomas 2007). First, sport programs can provide youth with opportunities to be physically active, which in turn can lead to improved *physical health*. Second, youth sport programs have long been considered important to youth's *psychosocial development*, providing opportunities to learn important life skills such as cooperation, discipline, leadership, and self-control. Third, youth sport programs are critical for the learning of *motor skills*; these motor skills serve as a foundation for recreational adult sport participants as well as future national sport stars.

Currently, youth sport programs are not producing outstanding results in any of these three objective areas. For example, child obesity rates in developed nations are high (e.g. Tremblay *et al.* 2002), while problem behaviors such as drug use and delinquency are on the rise among youth (e.g. Health Canada 2004). Further, attrition rates from youth sport programs are extremely high during adolescence, with an estimated one-third of all participants between 10 and 17 years of age withdrawing from sport programs every year (Gould 1987), leading sport psychology researchers to identify dropout as an area of concern (Brustad *et al.* 2001). In this chapter, we outline some of the current youth sport research and discuss youth sport participation within a broader integrated developmental and ecological model.

The developmental model of sport participation

Côté and colleagues' Developmental Model of Sport Participation (DMSP: Côté 1999; Côté and Fraser-Thomas 2007; Côté and Hay 2002) provides a framework with pathways that support continued participation for all youth, health benefits, and psychosocial development through sport. The DMSP emerged from extensive retrospective interviews with athletes in a variety of sports (e.g. hockey, baseball, gymnastics, rowing, tennis, and triathlon) and proposes three possible sport participation trajectories: 1) recreational participation through sampling, 2) elite performance through sampling, and 3) elite performance through early

specialization. Two of these trajectories, recreational participation and elite performance through sampling, have the same foundation from ages 6 to 12. After the sampling years, sport participants may choose to either stay involved in sport at a recreational level (*recreational years*, age 13+) or embark on a path that focuses primarily on performance (*specializing years*, age 13–15; *investment years*, age 16+). These two trajectories have different outcomes in terms of performance, but similar psychosocial and physical health benefits. The third possible trajectory is elite performance through early specialization. While this path often leads to elite performance, it may also lead to some negative physical and psychosocial outcomes.

Trajectory 1: Recreational participation through sampling

During the sampling years (age 6–12), athletes participate in a variety of sports with the focus being primarily on deliberate play activities. Côté and colleagues (Côté and Hay 2002; Côté *et al.* 2003) define deliberate play activities in sport as those designed to maximize inherent enjoyment. These activities are regulated by flexible age-adapted rules, and are set up and monitored by children or an involved adult. Children typically modify rules to find a point where their game most resembles the actual sport, but still allows for play at their level. These years are considered essential building blocks for recreational sport participation. The recreational years (age 13+) are usually seen as an extension of the sampling years, with the primary goals being enjoyment and health. Activities involve more deliberate play than deliberate practice. Deliberate practice is defined as highly structured activity that requires effort, generates no immediate rewards, and is motivated by the goal of improving performance rather than inherent enjoyment (Ericsson *et al.* 1993). During the sampling and recreational years, coaches are primarily supportive and encouraging. The roles of the parents during the sampling and recreational years include introducing their children to sports, enrolling them in diverse activities, and providing them with necessary resources, equipment, and playing opportunities. Parents should also provide adequate levels of emotional, informational, and tangible support to maintain their children's involvement in sport throughout development (Côté 1999).

Trajectory 2: Elite performance through sampling

For youth interested in a more performance-oriented path, a second trajectory of the DMSP suggests that specialization begins around age 13, after the sampling years. The specializing years (age 13–15) are seen as a transitional stage to the investment years (age 16+). During the specializing years, youth engage in fewer activities, which are a mix of deliberate play and deliberate practice activities, while during the investment years, youth commit to only one activity, and engage primarily in deliberate practice. A recent review of retrospective studies of elite level athletes in various sports (Côté *et al.* 2007) show that this trajectory is a common pathway to elite performance for sports in which peak performance is

achieved after puberty. During both the specializing and investment years, a more reciprocal coach–athlete respect develops, with coaches' styles becoming more skill-oriented and technical. Parents become less involved, but provide more financial and emotional support by helping their children through challenges and obstacles. Essentially, parents progress from a leadership role during the sampling years to a following and supporting role during the specializing and investment years (Côté 1999).

Trajectory 3: Elite performance through early specialization

In sports where peak performance is achieved before puberty (e.g. women's gymnastics, figure skating), early specialization is often necessary to reach elite performance. This phenomenon is also being extended to a variety of other sports (namely baseball and basketball) as the attractiveness of post-secondary athletic scholarships become increasingly lucrative (Gould and Carson 2004). However, studies that examine the development of elite athletes in sports such as basketball and baseball do not support an early specialization trajectory towards elite performance (see Côté *et al.* 2007, for a review). The early specialization path is characterized by high volumes of deliberate practice and low amounts of deliberate play in a context that focuses on performance. Thus, elite performers in these 'early specialization' sports usually skip the sampling years, and consequently may experience some negative physical and psychosocial outcomes during this period. For example, 'early specializers' often experience overuse injuries, reduced sport enjoyment and are more likely to eventually drop out of the sport they originally specialized in. Wall and Côté (2007) showed that young elite dropout ice hockey players began off-ice training (for the purpose of improving hockey performance) at a younger age and invested significantly more hours per year in off-ice training at ages 12–13 than a group of invested young elite ice hockey players. These results, along with the results of other qualitative studies of dropout and burnout athletes (e.g. Carlson 1988; Gould *et al.* 1996), indicate that engaging in more sport-specific training activities at a young age may have negative implications for long-term sport participation.

When a child's overall development is considered, many environmental factors can also impact on an individual's sport participation trajectory. García Bengoechea and Johnson state that '[U]nderstanding the forces that facilitate sports participation may contribute to the development and implementation of effective programs that increase involvement in sport activities, thereby enhancing physical, social, and psychological wellness in children' (2001: 20). The environment, therefore, becomes an important and interesting component to explore when examining youth sport developmentally. Bronfenbrenner's ecological theory (1977, 1999) provides an integrated approach to studying the development of youth through sport participation.

Ecological systems theory

Ecological systems theory refers to the notion that human development and human behavior are the materialization of person–context interactions (Bronfenbrenner 1995). Bronfenbrenner (1977) first conceptualized four nested systems (or levels) within an ecological network: 1) microsystem, 2) mesosystem, 3) exosystem, and 4) macrosystem. The first level, or microsystem, is comprised of participants, a physical domain, a location, and/or a program of activities. The second level, or mesosystem, is based on interrelationships between two or more microsystems involving the developing person (i.e. relationship between the coach and child). The third level, or exosystem, does not include the developing person within it; this level is representative of situations that affect the setting containing the person in question (i.e. interaction between coach and administrator). Lastly, the macrosystem includes cultural and social forces that impact human development. These nested systems are in constant interaction with each other to result in specific developmental processes and outcomes.

Bronfenbrenner later consolidated these ideas to form two specific propositions (1999). Proposition 1 postulates that human development, particularly in its early phases, occurs through processes of complex, reciprocal interactions between an active human organism and persons, objects, and symbols in its immediate environment. These interactions, or 'proximal processes', must occur on a regular basis and over a long period of time. Examples of patterns of these proximal processes include group or solitary play, reading, learning new skills, and athletic activities. Proposition 2 expands upon the nature of the proximal processes. This proposition states that the form, power, content, and direction of the proximal processes affecting development vary systematically as a joint function of: 1) the characteristics of the developing person, 2) the environment in which the processes are happening, 3) the developmental outcomes under consideration, and 4) the changes occurring over the time period in which the processes are taking place. This approach is aptly named the process–person–context–time (PPCT) model.

Linking the PPCT model to sport development

In this section we use the PPCT model to serve as a framework to further enhance understanding of healthy development through sport. In particular, components of the PPCT are integrated with concepts from the DMSP.

Process

It has been suggested that proximal processes are a critical catalyst for human development, acting as mechanisms of organism–environment interaction (Bronfenbrenner 1995). We suggest two proximal processes principles to promote positive youth development in sport, in accordance with the DMSP. First, we suggest a progression from deliberate play activities during childhood to progressively more deliberate practice activities during adolescence. Deliberate play

activities during childhood will help children become more self-directed towards their participation in sport (Ryan and Deci 2000; Vallerand 2001). Furthermore, deliberate play situations allow children the freedom to experiment with different movements and tactics, and the opportunity to learn to innovate, improvise, and respond strategically. Further, deliberate play activities in sport allow children to perfect skills that would not be practiced in organized situations (Côté *et al.* 2007). However, during adolescence it becomes more important to integrate deliberate practice activities into regular scheduling, particularly in elite sport programs.

Second, we suggest that children have the opportunity to sample various sporting activities. Sampling during childhood allows individuals to experience the context of various sports. For example, the proximal processes involved in basketball, soccer, tennis, golf, and track and field have distinctive features that can potentially lead to very different sport experiences and outcomes. We propose that sampling various sports allows individuals to establish different relationships with coaches, teammates, and other adults that have the potential to positively shape the course of development of youth.

Person

The second component of the PPCT focuses on the person as both a producer and product of his or her environment; this component of the model draws attention to the variation in characteristics of individuals involved in sport. Characteristics such as self-perceptions and motivation have been examined extensively in youth sport research (Horn 2004; Weiss and Williams 2004). Fraser-Thomas *et al.* (2005) integrate concepts of the DMSP related to the 'person' with Benson's (1997) forty developmental assets. The assets fall into two broad categories: external and internal. Within each of these categories, four types of assets exist. External assets include: support, empowerment, boundaries and expectations, and constructive use of time. Internal assets reflect an individual's values and beliefs and include commitment to learning, positive values, social competencies, and positive identity. The forty developmental assets are commonly termed the 'building blocks' for human development (Benson 1997). The more developmental assets an adolescent possesses, the greater his or her likelihood of developing in a positive and healthy manner (Benson *et al.* 1998; Scales and Leffert 1999). Although no single activity can promote all developmental assets, Fraser-Thomas *et al.* (2005) recently contended that participation in sport can be an important activity for the acquisition of several developmental assets.

Despite these claims, it has been suggested that sport is not necessarily the 'magic ingredient' to the further development of positive youth, adding that sport and other leisure activities are capable of having both positive and negative influences on development of an individual (Danish *et al.* 2004). In fact, Mahoney and Stattin (2000), in their examination of adolescents' low-structured and high-structured sport and leisure activities, found that it was the structure and context of the activities rather than the activity itself that determined whether the outcomes were positive or negative in terms of adolescent development. Programs that

focus on developmental assets integrate a broad range of individuals' internal and external attributes and increase the likelihood of healthy development through sport. Therefore, sport has the potential to contribute to a person's positive developmental outcomes if delivered within an appropriate framework such as the developmental assets framework.

Context

Programs must also be delivered in an appropriate context, the third component of the PPCT model. This component refers not only to the developing person's physical environment but also to the individuals within that environment who form strong bonds and relationships with the developing person (Bronfenbrenner 1999). After examining the impact of the physical environment on youth, the National Research Council Institute of Medicine (NRCIM 2002, 2004) suggested eight main features that should be present in the context of community programs in order to facilitate positive youth development. These features are gaining increasing support from youth sport research (e.g. Fraser-Thomas *et al.* 2005) as they offer additional understanding of the context in which youth sport should be promoted.

1. Physical and psychological safety

Physical and psychological safety in youth sport settings refers to the existence of safe and healthy facilities and practices that encourage secure and respectful peer interactions. Research indicates that the athlete–peer microsystem has an impact on the child's sense of physical self-worth (Vazou *et al.* 2006) and on the adolescent's perceived competence and self-evaluations (Horn 2004). Therefore, it is important that peer interactions are respectful in sport in order to build confidence in youth and allow them to enjoy their participation in sport.

2. Appropriate structure

This feature suggests the existence of clear and consistent expectations regarding rules and boundaries. The DMSP provides some guidelines for the structure of youth sport programs (e.g. a shift in focus from deliberate play during childhood to deliberate practice during adolescence). Thus one could contend that providing activities that are properly structured has the potential to develop positive, well-adjusted, and optimistic youth.

3. Supportive relationships

The third setting feature relates to strong support, positive communication, and connectedness. A coach can influence a child's perceived competence, enjoyment, and motivation (Black and Weiss 1992) and play a role in a child's psychological, social, and physical growth (Conroy and Coatsworth 2006; Côté

and Fraser-Thomas 2007). Training coaches about basic principles of positive youth development is likely to result into better youth sport programs and sporting environments that promote supportive relationships (Conroy and Coatsworth 2006; Smoll and Smith 1996).

4. Opportunities to belong

The fourth setting feature highlights the importance of meaningful inclusion, social engagement, and cultural competence in youth sport programs. Feeling a sense of belonging (i.e. being part of a team, developing friendships) is important in maintaining a child's motivation and interest in sport (Allen 2003). Healthy relationships can be encouraged by coaches who build a sense of team unity and cohesion.

5. Positive social norms

This feature relates to the development of values and morals rather than antisocial and reckless behaviors. Although a growing body of literature highlights some of the potential negative social norms associated with youth sport participation (e.g. violence, aggression, poor sportspersonship, and low morality reasoning: Bredemeier 1995; Lemyre *et al.* 2002), youth sport programs have the potential to develop positive values such as fair play, sportspersonship, cooperation, assertion, responsibility, empathy, and self-control (Côté 2002).

6. Support of efficacy and mattering

The sixth setting feature focuses on the importance of empowering youth and supporting their autonomy as they work to build their community. Research in sport emphasizes the need for coaches to develop autonomous athletes; giving youth the opportunity to choose their level of involvement in sport or contribution within a sport will empower them and also increase their intrinsic motivation for sport (Mallett 2005; Vallerand and Rousseau 2001).

7. Opportunities for skill building

The seventh setting feature emphasizes the importance of learning experiences. As previously outlined, sampling a variety of different sports through early diversification provides this opportunity, as youth have the chance to learn a variety of sport skills and are able to meet and interact with a variety of different people (i.e. peers, coaches). Furthermore, deliberate play and deliberate practice activities afford children and adolescents the opportunity to grow and develop their motor skills in appropriate settings.

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8. Integration of family, school, and community efforts

This feature promotes the melding of the young person's environments to increase communication and lessen conflicts and dissonance. In youth sport, parents play a key role in athletes' development of other supportive relationships, such as coach–athlete interactions (Jowett and Timson-Katchis 2005). Further, the structure and environment of a community appear to play a role in youth's persistence and progression in sport, given research suggesting that smaller cities tend to produce more professional athletes (Côté *et al.* 2007).

Time

Time is the last and often neglected component of the PPCT model. According to García Bengoechea and Johnson (2001), human development can only be fully understood if it is examined over an extended period of time. Thus, in order to truly comprehend child development in sport, individual attributes and their environmental interactions must be studied over time. The DMSP outlines how different activities benefit children's development at different ages, and provides a temporal progression of sport involvement with stages that are qualitatively and quantitatively different from each other. The first two trajectories (i.e. recreational participation through sampling and elite performance through sampling) include activities and environments that are similar during childhood and progressively different during early adolescence, late adolescence, and adulthood. For example, both trajectories include the sampling years, which focus on letting children experiment with various ways of executing sport skills in various contexts through deliberate play and involvement for fun in several sports. The DMSP suggest that children should spend more time in deliberate play activities than in deliberate practice activities during the sampling years (age 6–12). However, as children age and mature, the recommended amount of deliberate play activities can be slowly replaced by deliberate practice activities, depending on the eventual goals of the individual (e.g. recreation or elite sport participation). Thus, in youth sport programs, the time component of the PPCT model should focus on how playing and training activities change throughout development, as outlined by the DMSP.

Conclusion

This chapter has examined research in youth sport through an integrated developmental and ecological lens. The PPCT components of Bronfebrenner's ecological system theory and principles of positive youth development combined with the DMSP served as a framework to increase understanding of healthy youth development through sport participation. Based on youth sport research, particularly Côté and colleagues' DMSP, the following suggestions are made for youth sport programs aiming to promote positive youth development.

1 Sport programs for children (i.e. age 6–12) should include interactions between children, and between children and adults, that are based on play and opportunities to try out different forms of sporting activities. 'Sampling' and 'playing' during childhood are posited as the *proximal processes* that form the primary mechanism for continued sport participation at a recreational or elite level.
2 Sport programs during adolescence (i.e. age 13+) can change to include proximal processes built upon more specific training activities and specialization in one sport. As such, adolescents should have the opportunity to either choose to specialize in their favorite sport or continue in sport at a recreational level.
3 The developmental assets of the *person* (i.e. child or adolescent) involved in a sport program should be a priority of coaches, parents, and adults involved in the sport experience.
4 The eight setting features of the NRCIM should be implemented in sport programs to provide youth with a *context* that promotes developmental assets and the growth of life skills, competency, and responsibility.
5 Youth sport programs must be designed in consideration of children's healthy development over *time*. Administrators, coaches, and parents must look beyond the next game or the season final, to focus as well on the long-term positive developmental outcomes of the child-athlete.
6 The role of coaches and parents in sport is more than simply promoting motor skill development. Parents and coaches have a significant impact on the personal and social development of children involved in sport. Given the importance of these relationships, appropriate training that includes the principles of positive youth development should be provided to all adult leaders in youth sport programs.

The PPCT model serves to highlight the multiple features that should be combined to design and deliver youth sport programs that promote physical health, motor skill development, and psychosocial development. This chapter has showed that these three objectives are not mutually exclusive and that effectively designed sport programs can contribute to healthy youth development. In light of the role that sport can have on the whole development of youth, it becomes imperative that youth sport and positive youth development research inform each other on the best available means to promote enhanced life for all youth.

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